THEJAS DIXIT SATHYANARAYANA

Engineer – Robotics and Automation | Simulation & Application | Systems and Mechanical Engineering

Straubing, Germany | +49 15781288410 | thejasds21@gmail.com

linkedin.com/in/thejas-dixit-s | github.com/Thejas12Dixit | My Portfolio

PROFILE

Robotics and automation engineer with a research and applied background in robot simulation, offline programming, virtual commissioning, and mechanical design. M.Sc. in Systems Engineering for Manufacturing (OvGU Magdeburg). Research experience in PPR-based digital factory planning, bin-picking simulation, Industry 4.0 data standards (AutomationML, AAS, OPC-UA), and multi-brand OLP for KUKA, ABB, FANUC, Stäubli, and UR. EU Blue Card holder; available from April 2026.

PROFESSIONAL EXPERIENCE

Simulation Engineer, Robotics - Strama-MPS GmbH, Straubing
(10/2024 – 03/2026)

- Robot motion planning, collision detection, offline programming, and safe robot operation.
- On-site robot commissioning using KRL in accordance with BMW standards.
- Robot simulation for cycle-time optimisation across more than 15 projects.
- Build-up of simulation cells in Siemens Process Simulate with reachability and cycle-time analyses; evaluation of real plants against simulation results.
- Technical coordination with suppliers and end customers in body-in-white, e-mobility, and machining-centre applications; preparation of reports based on simulation results.

Intern & Master's Thesis - Volkswagen AG, Wolfsburg
(03/2023 – 07/2023)

- Master's thesis on the optimisation of an industrial bin-picking process using the lean simulation methodology and the PPR approach, with Universal Robots, Python, RoboDK, and CAD.

HiWi – Student Research Assistant - Otto von Guericke University, Magdeburg
(12/2022 – 03/2023)

- Research in cooperation with Volkswagen AG on robot programming and simulation.

Intern - Otto von Guericke University, Magdeburg
(10/2022 – 03/2023)

- Implementation of an Industry 4.0-based control system using PLC programming (IEC 61131) on WAGO PLC modules, an OPC-UA concept, and the AutomationML tool.

Jr. Design Engineer - Envision Integrated Services Pvt. Ltd., India
(08/2020 – 01/2021)

- Design projects with CATIA, Autodesk Inventor, Navisworks, and AutoCAD.
- Digital factory conversion and CATIA surfacing for automotive applications.

SKILLS AND KNOWLEDGE

- **Robotics and Simulation:**
  Robot simulation (KUKA, ABB, FANUC, Stäubli, UR), robot commissioning, Siemens Process Simulate, KUKA WorkVisual, RoboGuide, RobotStudio, Stäubli Robotics Suite, RoboDK, digital twin, virtual commissioning, Industry 4.0, ROS2 (in progress).
- **CAD:** SolidWorks, CATIA V5, Autodesk Inventor, Creo, Solid Edge, and AutoCAD.
- **Programming & Automation:**
  Python, KUKA KRL, PLC programming (IEC 61131), WAGO, e-Cockpit, Siemens S7 TIA.
- **Systems Engineering:** UML, SysML, AAS, Enterprise Architect, AutomationML, PLM, PTC Windchill, MBSE.

ACADEMIC QUALIFICATIONS

Master of Science in Systems Engineering for Manufacturing - Otto von Guericke University, Magdeburg (10/2021 – 02/2024)

- Focus: Model-based systems engineering, digital twin, factory automation, and industrial robotics.

Bachelor of Engineering in Mechanical Engineering - Visvesvaraya Technological University, Belagavi, India (07/2015 – 07/2019)

- Focus: Mechanical design, analysis, mechatronics, and automation.

CERTIFICATIONS

- KUKA Robot Programming I – KSS 8.x (KR C4 / KR C5) – (Basics) – (01/2025)
- KUKA Robot Programming II – KSS 8.x (KR C4 / KR C5) – (Advanced) – (02/2025)
- Safety Commissioning – Safe Operation Technology KR C4 / KR C5 – (03/2025)
- Universal Robot Programming – e-Series Core Track – (05/2023)

PROJECTS

- KUKA KRL Reader: Python application for visualising KUKA programs. Available on GitHub. (2026)
- Master's thesis: PPR-based industrial planning for lean robot operations. Backend simulation with RoboDK and Python for collision detection and reachability analysis. User interface for 3D visualisation of bin-picking models. (2023)
- Optimisation of an industrial bin-picking robot: gripper concept and motion planning. (2023)
- Development of collaborative robot-based industrial operating concepts with Universal Robots. (2023)
- Implementation of an Industry 4.0-based control system with cyber-physical systems and Asset Administration Shells. (2022–2023)
- Requirements-based design development for a zero-gravity aircraft concept in accordance with German VDI standards. (2021–2022)
- Bachelor's thesis: Generation of electricity and pyrolysis oil from plastic waste. (2018–2019)
- Design and development of a hydro-pneumatic integrated moulding-machine circuit. (2018)
- Presentation of a technical paper on: Biomimetics – Prosthetics Technology. (2019)

ADDITIONAL EXPERIENCE

- Earlier engineering internships, India (2018–2020): Product design and GD&T (Gloinnt Solutions) · Industrial hydraulic and pneumatic systems (Rexroth Bosch Lab, DSU) · Manufacturing processes and R&D in rail-vehicle production (BEML).
- Certified training programme in 3D design and printing technology. (2017)
- Finalist at the ISIE Indian Karting Championship (IKR 2016–2017), go-kart design and development.

LANGUAGES & INTERESTS

- **Languages:**
  - English (professional proficiency – C1),
  - German (fluent – B2),
  - Hindi (fluent),
  - Kannada (native).
- **Interests/Hobbies:** Travelling, cooking, hiking, football, reading, and photography.

Straubing, Germany

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